

# Benjamin Curis-Friedman

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## EDUCATION

### McGill University

*Bachelor of Software Engineering COOP – Minor of Mathematics* 3.96 GPA

Montreal, QC

Aug. 2024 – Dec. 2027

Awards (6000\$): Alma Mater Entrance Scholarship, Class of Engineering '83 Scholarship

### Stanford Online

*Machine Learning Specialization by Andrew Ng.* 100% Grade

May 2025 – August 2025

Covering regression, deep-learning, random forests, and ML engineering practices.

## EXPERIENCE

### Software Developer Intern

Jan. 2026 – Aug. 2026

*Rogue Research Inc*

Montreal, QC

- Coded vertex shaders in OpenGL to enable realistic optical distortion of a scene.
- Contributed to open source libraries (pyvista & vtk), improving code coverage and rendering speed by 10x.
- Engineered a 3D robot emulator application using pyvista, enabling accurate surgery planning.
- Refactored a brain surgery robot networking protocol to make it resilient to corrupt data packets.
- Implemented two robotic brain stimulation papers, leveraging bayesian optimization using Pytorch.
- Developed unit tests for existing classes and methods to improve code coverage in Objective-C.
- Implemented a vectorized SSIM image comparison algorithm for CI UI Tests using Apple's vImage framework.

### Artificial Intelligence Intern

May 2025 – Dec. 2025

*iSMART AI Lab*

Montreal, QC

- Developed a video synchronization & compression library using FFmpeg and OpenCV to prepare for training.
- Coded a data pre-processing pipeline to clean raw human vital signals for machine learning models.
- Built and programmed a task labeling device using Arduino and C++ to split recording segments efficiently.
- Created a high quality multi-modal dataset of over 50 hours of recordings from dozens of study participants.

## PROJECTS

### Omniclave 🌀 | *FastMCP, FastAPI, Auth0, React, Bun, Codex*

May 2026

- Built an MCP server for a student portal, empowering agents to help with homework and emails.
- Integrated OAuth2 to let users login from their favourite MCP client (i.e. Codex / Claude code).
- Automated a secondary login flow using a backend playwright instance to log into Omnivox on a user's behalf.
- Reverse engineered the Omnivox website to replicate the http headers and intercept the bearer token.

### AI Library From Scratch 🌀 | *Python, NumPy, TensorFlow, Keras, PyGame*

Jul. 2025

- Created a neural network library complete with mini-batch gradient descent and activation/cost functions.
- Trained a sequential neural network achieving ~96% test accuracy on MNIST dataset.
- Implemented customizable network and layer shapes for enhanced flexibility and scalability.
- Visualized inference with an interactive real-time digit recognition game using PyGame.

### Cheese Manager 🌀 | *Java, JavaFX*

Sept. 2025

- Developed an application for a comté cheese distribution business following the MVC pattern in Java.
- Created an aesthetically pleasing user interface with JavaFX using reusable components.
- Collaborated with teammates to create a UML class diagram and state diagram using Umlple.

### WebChat 🌀 | *Firebase, Javascript, Vue.js, HTML, CSS*

Sept. 2023

- Developed a full-stack web messaging platform with authentication to message friends.
- Programmed search and adding friends features with Google Cloud Functions.
- Implemented Firestore security rules to ensure secure communications.

## SKILLS

**Languages:** Python, C, Objective-C, Java, Bash, C++, Typescript & JavaScript, HTML, CSS, R

**Frameworks:** React, Node.js, JUnit, Tailwind, Cocoa, JavaFX

**Developer Tools:** Git, Github Actions, Docker, Firebase, CDash, VS Code, XCode, NeoVim, Linux, Subversion

**Libraries:** NumPy, Matplotlib, PyVista, VTK, OpenCV, pandas, Ax Platforms, SciPy, Keras, TensorFlow

**Spoken Languages:** English, French